Pooling and Drift in Delayed Bandits

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Abstract

A system often has to act long before it learns whether the act worked: a recommender sees a click in seconds and a purchase in days. For K actions and a delay of d rounds, the sharpest published rates are minimax-optimal in the delay when only the action is visible (Zimmert and Seldin, 2020), and $\widetilde{O}(\sqrt{(K+d)T})$ over T rounds once an intermediate state is (Esposito et al., 2023). Both are governed by the number of actions, so a longer menu is always more expensive to learn from, and neither adapts to instances in which many actions are effectively interchangeable. They need not be. If the outcome depends on the action only through the state it produced, a click or a browse or a cart, then one late outcome informs every action that could have produced the state observed, and the price is set by how many genuinely different states the actions produce. We charge each delayed outcome through the state and pay an *effective dimension* v_t, between 1 and the number of states, in place of K. For the single-copy algorithm that is actually run we prove $\widetilde{O}(\sqrt{V^-} + \sqrt{dT})$ for any budget fixed in advance, so the action count leaves the leading term and the delay enters additively. The gain has a limit: even when handed the exact losses from d rounds ago, no algorithm escapes $\Omega(\sqrt{d\mathcal{E}\min\{1+\log J, T/d\}})$, where J counts the drifting directions and $\mathcal{E}$ bounds how far the losses move while the learner waits. On generated data the state channel cuts regret by up to 79 per cent against action-level weighting, and on the funnel family by 32 to 68 per cent against Zimmert and Seldin (2020) under the tuning protocol we report. The guarantees assume the action-to-state map is known, and every experiment is synthetic.

1 Introduction

Operations commit before they learn. A markdown is set and sell-through is known a season later, staff are rostered and the shift's outcome lands after it, marketing spend is committed before attribution resolves, an item is recommended and the purchase it may cause is days away. In each the decision is made now and the quantity it will be judged by arrives late, while something cheaper arrives quickly: a fitting-room visit, a shift's first hour, a click. Whether that early signal can cut the cost of learning is a question between two fields. What may be inferred from the signal is a matter of data; the wait before the outcome resolves is a constraint on the operation, which no estimator removes, and the decision commits while only the first of the two is settled.

The cost of learning under delay should be set by how many genuinely different regimes the actions induce rather than by how many actions there are: a catalogue of thousands of items that funnels users through half a dozen engagement patterns should be no harder to learn from than a catalogue of six.

The mechanism that makes it true is that the outcome depends on the action only through the intermediate state it produced, so one late outcome can be charged to every action that could have produced the state observed rather than to the one action played. Bandits with intermediate observa-

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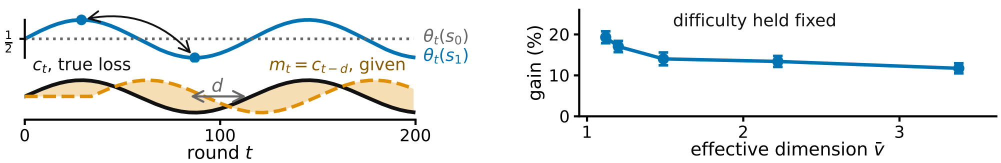

Figure 1: **The claim and its boundary, on generated data. Left,** what each state predicts changes while the action-to-state map stays fixed, so losses from d rounds earlier grow inaccurate; E_2 of Section 2 tracks it. **Right,** more overlap allows more sharing, at matched difficulty, $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$, 20 paired seeds, bars one standard error of the paired difference (Appendix E, study 10).

tions formalize the setting: the learner picks one action and sees the consequence of that action alone. Esposito et al. (2023) show that the *state-to-loss* map decides the complexity, an unrestricted one giving only the rate available with no intermediate observations; we fix that map and ask what the *action-to-state* structure buys. The delayed-feedback literature has since developed along other axes: Bar-On and Mansour (2025) obtain instance-dependent bounds when the observations themselves evolve, and Levy et al. (2025) treat contextual actions under delay with general function approximation. In neither does an intermediate state carry one delayed outcome across several actions.

What the pooling costs is an *effective dimension* v_t, measuring how distinguishable the action-induced state distributions are under the current play; it lies between 1 and the number of states and can fall far below K. Theorem 1 pays v_t in place of K for the algorithm that is actually run, and across the families the measured gain tracks v_t rather than K. The claim has a boundary. Waiting carries a cost of its own, which sharing cannot remove. Writing $m_t = c_{t-d}$ for the action-loss vector from d rounds earlier and $E_2 = \sum_t \|c_t - m_t\|_\infty^2$ for how far the losses move while the learner waits, Theorem 2 bounds below the regret this forces even when m_t is supplied free (Figure 1). Adapting to overlap and drift at once remains open (Section 5). Four ingredients separate the neighbouring settings, and it is their combination that is new here.

	delayed	state seen	P known	instance-dependent
Zimmert and Seldin (2020)	✓	✗	✗	✗
Esposito et al. (2023)	✓	✓	✗	✗
Eldowa et al. (2024)	✗	✓	✓	✓
Bar-On and Mansour (2025)	✓	✗	✗	✓
this paper	✓	✓	✓	✓

Where this work sits: the four ingredients that separate the neighbouring settings.

Eldowa et al. (2024) is closest, and $v_t - 1$ is exactly the χ^2 mutual information they introduce, so that quantity is theirs; what is new is its role once the outcome is delayed. Other neighbouring settings are placed in Appendix C.

2 Problem formulation

The question is whether the intermediate state can lower the cost of learning below what K alone forces. The model that makes it precise is the following. There are T rounds, K actions, a finite state space $\mathcal{S}$, a known action-to-state matrix P, and a fixed known delay d; all logarithms are natural and $\widetilde{O}$ hides logarithmic factors. Every round follows the chain $A_t \xrightarrow{P} S_t \xrightarrow{\theta_t} X_t$: the learner draws A_t from a distribution x_t, observes $S_t \sim P(\cdot \mid A_t)$ at once, and receives the delayed outcome only at time $t + d$, encoded as a loss $X_t \in [0, 1]$ so that smaller is better, with $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$, where $\mathcal{F}_{t-1}$ is everything seen before round t; outcomes are conditionally independent across rounds given the states. The conditional mean depends on A_t only through S_t. This *state-sufficiency* assumption licenses cross-action pooling: an outcome observed at state S_t may update every action that could have produced that state. If an action affects the outcome directly, in a way not represented in S_t, the argument no longer applies. Here x_t depends only on $\mathcal{F}_{t-1}$; X_r may first affect the action distribution at round $r + d + 1$, equivalently the round-t decision uses outcomes only from rounds $r \le t - d - 1$.

The matrix $P \in [0,1]^{K\times|\mathcal{S}|}$ is row-stochastic and fixed, and the *state-loss means* $\theta_t \in [0,1]^{|\mathcal{S}|}$ are *oblivious*: chosen in advance, not reacting to the learner. Action losses are $c_t = P\theta_t$ and performance is the static *pseudo-regret*, the gap to the best single action in hindsight, $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$. **Oracle for the lower bound.** For Theorem 2 only, the learner also receives the *stale action-loss vector* $m_t \triangleq c_{t-d}$ before choosing A_t, with $m_t = c_1$ for $t \le d$. This is not a function of the observables, so a lower bound surviving it is stronger, and STATE-EXP3 never uses it. We measure how misleading m_t is by $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \le T$, which stays small when errors are small even if frequent. The geometry of P governs sharing across actions and the evolution of θ_t governs staleness across time.

3 Pooling through the state

Two results follow: the number of actions stops setting the price for the algorithm one would actually run, and no sharing removes the cost of waiting.

An arriving outcome need not update the action played alone. Every action can be credited by how likely it was to have produced the observed state. Here P is known and m_t is not given. For each state $s \in \mathcal{S}$ let $q_t(s) = \sum_{a=1}^{K} x_t(a)P(s \mid a)$, and give each action $a \in \{1,\dots,K\}$ the estimate $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$. It is correct on average, and its noise is governed by the *effective dimension* $v_t = \sum_s q_t(s)^{-1} \sum_a x_t(a)P(s \mid a)^2 \in [1, |\mathcal{S}|]$. STATE-EXP3 is EXP3 (Auer et al., 2002) at learning rate η on these estimates: it adds $\hat{c}_t$ to its running total once the outcome is usable, at round $t+d+1$. A round costs $O(K)$ (Appendix A, with pseudocode and the χ^2 reading of v_t).

Theorem 1. *Let P be known and (θ_t) oblivious, and fix in advance any V^- with $\sum_t(v_t - 1) \le V^-$ always. Then* STATE-EXP3 *with any $\eta \le 1/(e(d+1))$ satisfies $\mathbb{E}[R_T] \le \sqrt{2(3.3V^- + 2dT)\log K} + e(d+1)\log K + d$.*

So K is replaced by V^-, which counts how differently the actions behave rather than how many there are, and the delay enters additively at $\widetilde{O}(\sqrt{V^-} + \sqrt{dT})$, for the algorithm every experiment below runs. Writing $\bar{v}$ for the largest v_t over all ways of playing, $V^- = (\bar{v} - 1)T$ is admissible and depends on P alone. A rotating variant that interleaves $d+1$ copies, and a version that merges similar states, are deferred to Appendix B; neither adapts to E_2.

The proof turns on one pathwise identity. The denominator of $\hat{c}_t$ is the state marginal of the distribution supplying its numerator, so $\langle x_t, \hat{c}_t\rangle = X_t \le 1$ on every path, whatever $q_t(S_t)$ is. That bounds how far the play distribution moves while an outcome is in flight, which is what the delayed cross-term needs; centring by $X_t\mathbf{1}$ replaces v_t by $v_t - 1$ (Appendix F).

The cost of waiting. Waiting carries a cost that persists even when the stale losses are supplied free. Theorem 2 isolates that cost from time alone: the construction gives each of the J drifting directions a private state, which removes useful pooling and forces $J \le |\mathcal{S}| - 1$ (Appendix D).

Theorem 2. *Fix T, a delay $1 \le d \le T/2$, a drift budget $\mathcal{E} \le T/4$, and $1 \le J \le \min\{K-1, |\mathcal{S}|-1\}$. Every algorithm, even one handed the exact losses from d rounds ago, meets some instance fixed in advance with $E_2 \le \mathcal{E}$ on which $\mathbb{E}[R_T] = \Omega(\sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$. Without that help, and if $K \le T$, the best regret any algorithm can guarantee is $\Omega(\sqrt{KT} + \sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$, up to constants.*

The first term is the usual cost of exploring. The second is the cost of delay: it grows with the wait, how far the losses move during it, and the number J of directions the best fixed action can use. They come from different hard instances, so adding them overstates the truth by at most a factor of two.

4 Experiments

A claim about instances is settled where the two counts it separates, how many actions there are and how many distinct behaviours they have, come apart. The families below are built so they do, and Figure 2 shows the two counts coming apart in both directions: the gain rising as the effective dimension falls, and the effective dimension holding while the catalogue grows. With $K = 200$ generated items and six levels of engagement the estimated dimension is 1.97, and the state-pooled

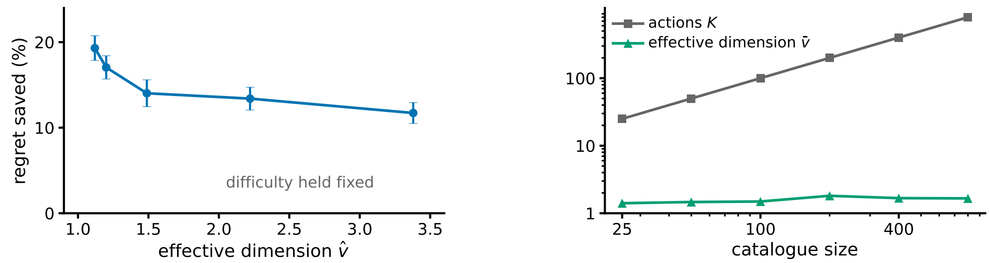

Figure 2: **Effective dimension, not action count. Left,** regret saved against action-level weighting as the effective dimension falls, with task difficulty held fixed so the sweep is not confounded by easier instances; $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$, 20 paired seeds, bars one standard error of the paired difference (study 10). **Right,** the catalogue grows thirty-twofold while $\bar{v}$ stays between 1.4 and 1.8, the separation Theorem 1 charges for (study 14/16). $\widehat{v}$ and $\bar{v}$ estimate $\sup_x v(x)$ by Monte Carlo.

variant cuts regret against action-level weighting by 79.3, 63.6 and 38.9 per cent at $d = 10, 50, 200$, and against the minimax-optimal delayed-bandit method of Zimmert and Seldin (2020) tuned over a grid by 67.9, 51.4 and 31.5. That setting is hand-structured, not fitted to data, so it shows the mechanism, not a deployed system.

d	action-level	tuned (Zimmert and Seldin, 2020)	STATE-EXP3	cut	cut
10	4577	2959	949	79.3%	67.9%
50	4752	3557	1728	63.6%	51.4%
200	5307	4735	3245	38.9%	31.5%

Funnel family, $K = 200$, $|\mathcal{S}| = 6$, $T = 20000$, 8 paired seeds; cuts are against action-level weighting and against the tuned baseline (study 14/16).

These runs use the single-copy algorithm that Theorem 1 covers; Appendix E also reports the rotating variant. A best-state rule leads seven of nine settings, but where its assumption fails its regret grows linearly in T where ours stays sublinear (Appendix E).

5 Discussion

Two things stand between an action and its worth: which actions resemble it, and how long the answer takes. The first is exploitable and checkable before deployment, since P is known and an estimate of $\bar{v}$ well below K says whether the state channel is worth using. The second is not: Theorem 2 shows the uncertainty left by waiting survives even when the exact losses from d rounds ago are handed over free. Two limits remain: $\bar{v}$ is estimated rather than solved for, and the bounds lie on different axes. Adapting to overlap and to drift at once, in one algorithm and one bound, is open.

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A The state-pooled estimator

The algorithm. Inputs are the action-to-state matrix P, the delay d, the number of copies m, and the rate $\eta > 0$. Setting $m = 1$ gives STATE-EXP3, the algorithm of the body and of every experiment, covered by Theorem 1; setting $m = d + 1$ gives a distinct algorithm, INTERLEAVED-STATE-EXP3, which runs $d + 1$ copies in rotation with copy $i \in \{0, \dots, m - 1\}$ taking every mth round, and is covered by Theorem 3. Round r's outcome becomes usable at round $r + d + 1$, which for $m = d + 1$ is the next round belonging to the copy that played round r.

1. Set $L_i \leftarrow (0, \dots, 0) \in \mathbb{R}^K$ for $i = 0, \dots, m - 1$, and let $\mathcal{P}$ be an empty table of rounds awaiting their outcome.
2. For $t = 1, \dots, T$:
   (a) *Deliver.* If $r \triangleq t - d - 1 \geq 1$, read $(i_r, q_r(S_r), S_r, X_r)$ from $\mathcal{P}$, remove it, and update only that copy, $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r \mid a) X_r / q_r(S_r)$ for every a. This is the pooled step, and it touches every action rather than the one played.
   (b) *Select the copy.* $i \leftarrow t \bmod m$.
   (c) *Play.* $x_t(a) \propto \exp(-\eta L_i(a))$, draw $A_t \sim x_t$, and observe S_t at once.
   (d) *Record.* Compute the scalar $q_t(S_t) = \sum_a x_t(a) P(S_t \mid a)$ and store $(i, q_t(S_t), S_t, \cdot)$ in $\mathcal{P}$, to be completed when X_t arrives at time $t + d$.

Steps (a), (c) and (d) are each $O(K)$ and there is no optimization subproblem, so a round costs $O(K)$ once P is stored. The state is held in the m vectors L_i, which is mK numbers, the matrix P, which is $K|\mathcal{S}|$ numbers, and four scalars for each of the at most d rounds in flight. Only the scalar $q_r(S_r)$ is needed from round r, not the vector x_r.

A worked example. Take three actions and two states, with

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \\ 0 & 1 \end{pmatrix}, \qquad x_t = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}),$$

so $q_t(s_1) = \frac{1}{2}(0.8) + \frac{1}{4}(0.4) + \frac{1}{4}(0) = 0.5$ and $q_t(s_2) = 0.5$. The third action cannot produce s_1 at all. Suppose the learner draws a_1, observes s_1, and receives $X_t = 1$ at time $t + d$. Then $\hat{c}_t(a) = P(s_1 \mid a)/0.5$, so $\hat{c}_t = (1.6,\, 0.8,\, 0)$ against the action-level $\tilde{c}_t = (2, 0, 0)$. One observation updates all three actions: a_2 is charged although it was never played, because it had a 0.4 chance of producing the state seen, and a_3 is charged nothing, because it could not have produced that state. Nothing is biased by this. With $\theta_t = (1, 0)$ the true losses are $c_t = (0.8, 0.4, 0)$, and averaging $\hat{c}_t$ over $S_t \sim q_t$ returns exactly that. What is bought is the second moment: $v_t = 1.44$ here, against $K = 3$ for the action-level estimate, and with $\theta_t = (1, 1)$ both are attained exactly.

Remark 1 (The χ^2 form). Under x_t the pair (A_t, S_t) has law $x_t(a)P(s \mid a)$ with marginals x_t and q_t, so

$$\begin{aligned} \chi^2\big(P_{A_t,S_t} \,\big\|\, P_{A_t} \otimes P_{S_t}\big) &= \sum_{a,s} \frac{\big(x_t(a)P(s \mid a) - x_t(a)q_t(s)\big)^2}{x_t(a)q_t(s)} \\ &= \sum_{a,s} \frac{x_t(a)P(s \mid a)^2}{q_t(s)} - 2 + 1 \;=\; v_t - 1, \end{aligned}$$

the cross term summing to $\sum_{a,s} x_t(a)P(s \mid a) = 1$ and the last to $\sum_{a,s} x_t(a)q_t(s) = 1$. So $v_t = 1$ exactly when A_t and S_t are independent, which pins the rows of P only on the support of x_t and therefore means all rows agree whenever x_t has full support, as exponential weights does. If P is deterministic and every state is reached by some action in the support of x_t, then v_t equals the number of states that support reaches. In particular, under full-support play and an onto deterministic P, $v_t = |\mathcal{S}|$. This is an identity, not a bound; it is used for reading v_t, and once in the proof of Theorem 1.

Proposition 1. *Fix t, let x_t be the play distribution, $q_t(s) = \sum_a x_t(a)P(s \mid a)$ the induced state distribution, and $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$, which is well defined because only states with $q_t(s) > 0$ occur. For statements involving $1/x_t(a)$, restrict to actions with $x_t(a) > 0$;* State-EXP3 *itself has full support on every action. Then $\hat{c}_t(a) \geq 0$, $\hat{c}_t(a) \leq 1/x_t(a)$, $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$, and*

$$\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \;\leq\; v_t := \sum_{s: q_t(s) > 0} \frac{\sum_a x_t(a)P(s \mid a)^2}{q_t(s)}, \qquad 1 \leq v_t \leq |\mathcal{S}|,$$

states of zero probability being omitted throughout, which is the convention $0/0 = 0$. The action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$ has the same first three properties and weighted second moment at most K.

Proof. Nonnegativity is immediate from $X_t \geq 0$. Given $\mathcal{F}_{t-1}$ the state S_t has distribution q_t, and $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$ by the model, since that conditional mean does not depend on the action, so $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = \sum_s q_t(s)P(s \mid a)\theta_t(s)/q_t(s) = c_t(a)$. For the range, $q_t(s) \geq x_t(a)P(s \mid a)$ for every a, so $P(s \mid a)/q_t(s) \leq 1/x_t(a)$ and $X_t \leq 1$. For the second moment, $X_t^2 \leq 1$ gives $\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq \sum_s P(s \mid a)^2/q_t(s)$, and exchanging the sums gives v_t. Each summand of v_t is at most 1, since $P(s \mid a) \leq 1$ makes $\sum_a x_t(a)P(s \mid a)^2 \leq q_t(s)$, and at least $q_t(s)$, since Jensen applied to $a \mapsto P(s \mid a)$ under x_t gives $\sum_a x_t(a)P(s \mid a)^2 \geq q_t(s)^2$. Summing the two bounds gives $1 \leq v_t \leq |\mathcal{S}|$. The action-level claims are the same computation with the state replaced by the played action, whose weighted second moment is $\sum_a x_t(a)/x_t(a) = K$. □

The two ends are attained. If every action induces the same state distribution then $v_t = \sum_s q_t(s) = 1$, and if P is deterministic, each action reaching a single state, then every $P(s \mid a)$ is 0 or 1 and each summand is exactly 1, so v_t counts the states reached and equals $|\mathcal{S}|$ when every state has an action reaching it. Disjoint row supports give $v_t = K$ rather than $|\mathcal{S}|$, since the summands over one action's private states add to $\sum_s P(s \mid a) = 1$, so a sensor with many private states per action resolves few of them. So v_t reads the overlap between the rows of P rather than their length, and

$\bar{v} = \sup_x v(x)$, the supremum over play distributions, depends on P alone. An outcome observed at a state updates the estimate for every action that can reach it, which is why the construction of Section 3 gives each drifting coordinate a private state, forcing $v_t = |\mathcal{S}|$ there. The immediately observed pairs (A_t, S_t) identify P separately, though Theorem 3 assumes it known.

B Proofs: pooling and grouping

Two results are stated here rather than in the body, since Theorem 1 covers the algorithm actually run. The first is the interleaved variant.

Theorem 3. *Let P be known and (θ_t) oblivious. Pick before the run any fixed number V with $\sum_t v_t \leq V$ always. Then* INTERLEAVED-STATE-EXP3 *at the matching learning rate satisfies $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V \log K}$. Writing $\bar{v}$ for the largest v_t over all ways of playing, $V = \bar{v}T \leq |\mathcal{S}|T$ is one such choice, and it depends on P alone.*

The second merges states that are not worth telling apart. Group them with a map g from $\mathcal{S}$ onto a smaller set $\mathcal{Z}$ and run the same algorithm on $g(S_t)$, where $P^g(z \mid a) = \sum_{s:g(s)=z} P(s \mid a)$. Grouping never raises the dimension, but the estimate now tracks a group average, so the widest spread of losses inside a group, $\delta_t(g) = \max_z \max_{s,s':g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$, is the error introduced.

Theorem 4. *Under the hypotheses of Theorem 3, running on g with $\sum_t v_t(g) \leq V(g)$ gives $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V(g) \log K} + 2\sum_t \delta_t(g)$.*

Merging need not preserve state-sufficiency, so the grouped estimate is right not for c_t but for a stand-in loss differing from it by at most $\delta_t(g)$ on every action. Keeping states apart preserves differences that matter but learns more slowly; merging shares more but pays $2\sum_t \delta_t(g)$. The best grouping is the true one: with 16 states from four hidden groups, sizes $1, 2, 4, 8, 16$ give regret 1590, 831, 754, 920, 1080, lowest at four, which no algorithm was told. The map g is fixed in advance; learning it from data remains open.

Proof idea. The pooled estimator is unbiased for every action, while Proposition 1 bounds its play-weighted second moment by v_t rather than K. With $m = d + 1$ interleaved copies, the outcome generated on a round belonging to one copy is available before that copy acts again, so each copy is an ordinary exponential-weights process with immediate feedback. Summing the $d + 1$ potential inequalities gives $(d + 1) \log K/\eta$, and the pooled second moments give $\frac{\eta}{2} \sum_t v_t$.

Proof of Theorem 3. Fix a copy i and let T_i be the number of rounds it owns, so $\sum_i T_i = T$. Round t's outcome is usable from round $t + d + 1$, which with $m = d + 1$ is exactly $t + m$, the copy's next round, so within a copy the feedback is undelayed and L_i carries exactly the estimates of that copy's earlier rounds. Exponential weights on nonnegative losses give, for any action $a^\star$ and any $\eta > 0$,

$$\sum_{t \in i} \big(\langle x_t, \hat{c}_t \rangle - \hat{c}_t(a^\star)\big) \leq \frac{\log K}{\eta} + \frac{\eta}{2} \sum_{t \in i} \sum_a x_t(a) \hat{c}_t(a)^2,$$

by the usual potential telescoping with $e^{-z} \leq 1 - z + z^2/2$, which needs $z \geq 0$ and therefore $\hat{c}_t \geq 0$, supplied by Proposition 1. Each x_t is $\mathcal{F}_{t-1}$-measurable, because L_i then holds only rounds $r \leq t - m = t - d - 1$, so taking expectations turns the left side into $\sum_{t \in i}(\langle x_t, c_t \rangle - c_t(a^\star))$ by unbiasedness and the quadratic term into at most $\mathbb{E}[\sum_{t \in i} v_t]$, again by Proposition 1. Summing over the $m = d+1$ copies and using $\sum_i \min_a \sum_{t \in i} c_t(a) \leq \min_a \sum_t c_t(a)$, which is what lets the copies be compared against one common action, gives $\mathbb{E}[R_T] \leq (d+1) \log K/\eta + \frac{\eta}{2}\mathbb{E}[\sum_t v_t]$. If $\sum_t v_t \leq V$ surely then $\eta = \sqrt{2(d+1) \log K/V}$ balances the two terms and gives $\sqrt{2(d+1)V \log K}$, and $V = \bar{v}T$ is admissible by the definition of $\bar{v}$. $\square$

The delay enters as a factor because the copies do not share their estimates, each paying $\log K/\eta$ over $T/(d + 1)$ rounds. Sharing them gives the $m = 1$ algorithm of Theorem 1, whose additive delay term avoids this penalty.

Proof of Theorem 4. Running the algorithm on g is running it on the sensor $(\mathcal{Z}, P^g)$. The surrogate losses below depend on x_t and so are predictable rather than oblivious, but the proof of Theorem 3 uses obliviousness nowhere, only that each c_t^g is $\mathcal{F}_{t-1}$-measurable given the play, so it applies unchanged and bounds regret measured in the surrogate losses $c_t^g(a) = \sum_z P^g(z \mid a)\bar{\theta}_t(z)$, where $\bar{\theta}_t(z) = \sum_{s:g(s)=z} q_t(s)\theta_t(s)/q_t^g(z)$ is the group mean the grouped estimator is unbiased for, by the same computation as in Proposition 1 with S_t replaced by $g(S_t)$. Fix a and z. Both $\{q_t(s)/q_t^g(z)\}$ and $\{P(s \mid a)/P^g(z \mid a)\}$ are probability vectors on $\{s : g(s) = z\}$, so their θ_t-averages lie in a common interval of length $\delta_t(g)$ and differ by at most $\delta_t(g)$. Weighting by $P^g(z \mid a)$ and summing over z gives $|c_t^g(a)-c_t(a)| \le \delta_t(g)$ for every a, hence $\langle x_t, c_t\rangle - c_t(a^\star) \le \langle x_t, c_t^g\rangle - c_t^g(a^\star) + 2\delta_t(g)$. Summing over t adds $2\sum_t \delta_t(g)$. Merging leaves $V(g)$ no larger than a bound valid for the raw states: it suffices to merge two states, the general case following by induction on the merges. Writing P_1, P_2 for the two columns, $B_j = \sum_a x_t(a)P_j(a)$ and $\lambda = B_1/(B_1 + B_2)$, Cauchy–Schwarz gives $(P_1 + P_2)^2 \le \lambda^{-1}P_1^2 + (1-\lambda)^{-1}P_2^2$ pointwise in a, so after dividing by the merged denominator the merged summand is at most the two it replaces. □

C Relation to other inaccuracy measures

Neighbouring settings. Vernade et al. (2020) vary the action-to-state map while fixing the state-to-loss map, where we do the reverse. Wang et al. (2026) and Zhang et al. (2026) measure a prediction error as E_2 does, without the delay. Feedback graphs (Alon et al., 2015; Esposito et al., 2022; Gabbianelli et al., 2023) reveal other actions' losses outright, where here the state's loss must still be estimated from the one outcome that arrives.

Write $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$ for the inaccuracy measure of Bar-On and Mansour (2025). Unlike E_2, which charges only the worst action in a round, Λ_2 grows with how many actions are predicted wrongly, and since $\|v\|_\infty \le \|v\|_2 \le \sqrt{J}\|v\|_\infty$ for a vector v with J nonzero coordinates,

$$E_2 \;\le\; \Lambda_2 \;\le\; J\,E_2,$$

with both ends attained: the left when a single action is mispredicted, the right when all J are mispredicted equally. So a bound in E_2 is never weaker and can be J times stronger, which is why Theorem 2 is stated in E_2. The two measures agree exactly when the prediction error is concentrated on one action.

D Proofs: the lower bound

The measure E_2 is controlled by the state-loss variation W, and no converse bound holds in general.

Lemma 1 (Delay window)**.** *$E_2 \le d^2W$ for $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$. The ratio $E_2/(d^2W)$ tends to one when θ drifts monotonically in one coordinate at a constant rate and $T/d \to \infty$, the first d rounds contributing $\sum_{j<d} j^2$ rather than d^2 each under the convention $m_t = c_1$.*

In words, accumulated squared prediction error is at most the state-loss variation inflated by the square of the delay, and no better bound in W alone is available.

Proof of Lemma 1. Throughout write $\Delta_j = \theta_j - \theta_{j-1}$, set $\theta_0 = \theta_1$, so $\Delta_1 = 0$, and read c_r as c_1 for $r \le 0$, matching the convention $m_t = c_1$ for $t \le d$ of Section 2. Then $c_t - c_{t-d} = P\sum_{j=t-d+1}^{t} \Delta_j$. Because the rows of P are probability vectors, averaging cannot increase the supremum norm, so $\|c_t - c_{t-d}\|_\infty \le \sum_j \|\Delta_j\|_\infty$. By Cauchy–Schwarz applied to the d summands, $\|c_t - c_{t-d}\|_\infty^2 \le d\sum_{j=t-d+1}^{t} \|\Delta_j\|_\infty^2$. Summing over t and using that each increment belongs to at most d windows gives $E_2 \le d^2W$. The ratio $E_2/(d^2W)$ approaches one when every increment has the same magnitude, sign and active coordinate, and some action reaches that coordinate deterministically, since then no cancellation occurs within a full window and P passes the increment through undiminished. Without the second condition the drift can lie in the kernel of P and the ratio can be zero. □

The construction in full. Take states $s_0, \dots, s_q$, a stationary deterministic map sending a_j to s_j for $j \le J$ and every other action to s_0, and independent Rademacher signs $\sigma_b^{(j)}$ for $2 \le b \le B = \lfloor T/h \rfloor$. Set $\theta_t(s_0) = \frac{1}{2}$ throughout, $\theta_t \equiv \frac{1}{2}$ on block 1, and $\theta_t(s_j) = \frac{1}{2} - \varepsilon\sigma_b^{(j)}$ on each later block b. The neutral first block makes $c_1 = \frac{1}{2}\mathbf{1}$, so the convention $m_t = c_1$ for $t \le d$ of Section 2 supplies a constant over exactly the rounds that precede any usable feedback, which is what the isolation step below needs at $b = 1$. Rounds after the last complete block carry no drift, costing a constant factor.

Lemma 2 (Rademacher lower tail)**.** *Let S_B be a sum of $B \geq 32$ independent Rademacher signs. For every $1 \leq u \leq \sqrt{B}/32$, $\Pr(S_B \geq u\sqrt{B}) \geq \frac{u}{4}e^{-64u^2}$.*

Proof. Write $S_B = 2X - B$ with $X \sim \mathrm{Bin}(B, \frac{1}{2})$, so the event is $\{X \geq B/2 + u\sqrt{B}/2\}$; passing to X removes the lattice gap, since X takes every integer value in $[0, B]$ whereas S_B takes only those of the parity of B. Put $n = \lfloor B/2 \rfloor$ and $t = \lceil u\sqrt{B}/2 \rceil + 1$, so $X \geq n+t$ implies the event and $t \leq u\sqrt{B}$. Consecutive binomial probabilities have ratio $(B - n - j + 1)/(n + j)$, whose distance from one is at most $8j/B$; for $j \leq B/16$ that is at most $\frac{1}{2}$, where $\log(1 - x) \geq -2x$ holds, so summing over $i \leq j$ gives $\Pr(X = n + j) \geq \Pr(X = n)e^{-16j^2/B}$ with $\Pr(X = n) \geq 1/(2\sqrt{B})$. Summing over the t integers $j \in \{t, \ldots, 2t - 1\}$, all at most $B/16$ because $u \leq \sqrt{B}/32$, and using $t \geq u\sqrt{B}/2$, gives $\Pr(X \geq n + t) \geq t\, e^{-64u^2}/(2\sqrt{B}) \geq \frac{u}{4}e^{-64u^2}$. $\square$

Three regimes give the maximal inequality the lower bound needs. Each $S_B^{(j)}$ is sub-Gaussian with variance proxy B, so $\mathbb{E}\max_j(S_B^{(j)})^+ \leq \sqrt{2B\log J} + \sqrt{B}$ throughout. For the matching lower bound, first suppose $\log J \leq B/8$ and $\log J \geq 128$. Taking $u = \sqrt{\log J/128}$, which then lies in $[1, \sqrt{B}/32]$, makes $e^{-64u^2} = J^{-1/2}$, so $\Pr(S_B^{(j)} \geq u\sqrt{B}) \geq \frac{u}{4}J^{-1/2}$ and independence across j gives $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \leq e^{-u\sqrt{J}/4}$, at most $\frac{1}{2}$ beyond an absolute J. Hence $\mathbb{E}\max_j(S_B^{(j)})^+ \geq \frac{1}{2}u\sqrt{B} = \Omega(\sqrt{B\log J})$. Second, if $\log J > B/8$ and $B \geq 1024$, take $u = \sqrt{B}/32$, so $u\sqrt{B} = B/32$ and $e^{-64u^2} = e^{-B/16}$, and $J \geq e^{B/8}$ makes $J\, e^{-B/16}$ diverge, giving $\mathbb{E}\max_j(S_B^{(j)})^+ = \Omega(B)$, which is $\Omega(\sqrt{B\min\{1 + \log J, B\}})$ in that regime. Third, when $\log J < 128$ or $B < 1024$ the quantity $\sqrt{B\min\{1 + \log J, B\}}$ is $O(\sqrt{B})$, and a single walk already gives $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \geq \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \geq \frac{1}{2}\sqrt{B/3}$ by Hölder, since $\mathbb{E}S_B^2 = B$ and $\mathbb{E}S_B^4 \leq 3B^2$. Combining the three regimes, $\mathbb{E}\max_j(S_B^{(j)})^+ = \Theta(\sqrt{B\min\{1 + \log J, B\}})$. This is the route by which maxima of independent random walks fix the minimax rate for prediction with expert advice (Cesa-Bianchi and Lugosi, 2006).

Proof idea. Divide time into blocks of length d and give J actions private states whose losses are perturbed by independent Rademacher signs within each block. Every observation available when an action is chosen comes from an earlier block, and the supplied $m_t = c_{t-d}$ depends only on earlier signs, so the learner cannot predict the signs governing its current block. Its expected loss is therefore $T/2$, while the best fixed action in hindsight benefits from the maximum of J independent random walks. Choosing the amplitude to make $E_2 \leq \mathcal{E}$ gives the stated scale.

Proof of Theorem 2. Take $h = d$ and $\varepsilon = \frac{1}{2}\sqrt{\mathcal{E}/T}$. Every outcome usable when choosing A_t comes from a round $r \leq t-d-1$, and $t \leq bh$ with $h \leq d$ forces $r < (b-1)h$, so A_t is independent of block b's own signs; the same computation places the stale vector m_t in blocks strictly before b, so handing it to the learner changes nothing. Every algorithm's expected loss therefore equals $T/2$, since its action is independent of the current block's signs and all states have mean $\frac{1}{2}$ marginally. Its regret is therefore exactly the comparator's fluctuation advantage, $\varepsilon d\, \mathbb{E}\big[\max_{j \leq J}\big(\sum_{2 \leq b \leq B}\sigma_b^{(j)}\big)^+\big]$ with $B = \lfloor T/d \rfloor$, over the $B-1$ randomized blocks. Per round $\|c_t - m_t\|_\infty \leq 2\varepsilon$, because the supremum norm charges only the largest coordinate and each coordinate moves by at most 2ε at a block boundary; the gap is ε across the first randomized block, where m_t is still the neutral c_1, and lies in $\{0, 2\varepsilon\}$ thereafter. Hence $E_2 \leq 4\varepsilon^2 T = \mathcal{E}$ holds deterministically. The budget must bind on the realized sequence, since a budget holding only in expectation would not constrain the instance selected by Yao's principle. By Lemma 2 and the discussion following it, the expected maximum of the positive parts of J independent Rademacher sums of length $B - 1$ is $\Theta(\sqrt{(B - 1)\min\{1 + \log J, B - 1\}})$, which covers both regimes and the case $J = 1$. The hypothesis $d \leq T/2$ gives $B \geq 2$ and hence $B - 1 \geq B/2$, so replacing $B - 1$ by B costs a factor at most $\sqrt{2}$ in each branch of the minimum, including the saturated branch where the minimum is $B - 1$ rather than $1 + \log J$. Substituting $B = \lfloor T/d \rfloor$ and ε therefore gives $\Theta(\sqrt{d\mathcal{E}\min\{1 + \log J, T/d\}})$. Rounds after the last complete block carry no drift and change the constant only. When the oracle is withheld, the $\sqrt{KT}$ term is

Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present model class. □

Remark 2 (Where the bound is matched). Suppose $X_t = c_t(A_t)$. On instances whose drift is carried by a single action only one coordinate of $c_t - m_t$ is nonzero, so $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$; on the family of Section 3 that coordinate has magnitude at most $2\varepsilon \le 1$, so the truncation at one in Λ_2 is inactive and $\Lambda_2 = E_2$. The bound of Bar-On and Mansour (2025) then reads $\widetilde{O}(\sqrt{KT + d\,E_2})$, whose $\sqrt{d\,E_2}$ contribution Theorem 2 matches up to the logarithmic factor at $J = 1$, so on that family the drift term of the published upper bound is unimprovable in order. Its $\sqrt{KT}$ term is not matched, the oracle supplying an entire stale loss vector, so the oracle-assisted problem need not retain bandit exploration complexity. Neither result provides an E_2-adaptive upper bound under noisy outcomes.

E Numerical detail

Sixteen studies were run and the five bearing on the theorems are reported here, each with an observation, an interpretation and a limitation. They keep their original numbers, which index the released scripts, so the numbering is not contiguous; the code for all sixteen is available. Studies 1 to 13 draw P from a Dirichlet and 14 to 16 do not. All policies run on the same environment parameters and common random-number streams, so a seed fixes everything except which estimator is formed, and differences are taken within a seed before averaging. Round r's outcome is consumed before round $r{+}d{+}1$, matching Section 2. Reported $\pm$ is one standard error of the paired difference. *Sample-path pseudo-regret* means $\sum_t c_t(A_t) - \min_a \sum_t c_t(a)$ on one realization of the learner and environment randomness, computed from the true losses rather than the sampled outcomes; averaging it over seeds estimates the expected pseudo-regret the theorems bound.

The first three studies probe the state channel of the claim from increasingly controlled and increasingly operational angles, and the last probes its temporal boundary. Study 1 raises K with $|\mathcal{S}|$ fixed and asks whether pooling grows more valuable, which is Theorem 1's prediction; study 10 holds K and the task difficulty fixed and varies the overlap instead, which separates that prediction from the instances simply becoming easier; study 14/16 shows the regime in an environment built to the shape of a funnel and measures it against a rate-optimal baseline. Study 8 then tests Theorem 2 on the drift channel, where pooling cannot help. Theorem 3's bound on the interleaved algorithm is checked inside study 1. Theorem 4, on coarsening, has no study here; it is stated and proved in Appendix B and left untested in this submission. Studies whose conclusions were superseded by a later control, or that replicated one already reported, are left to the released code.

Study 1. The gain grows with K at fixed $|\mathcal{S}|$. With $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, drift band 0.05 and 100 seeds:

K	interleaved	STATE-EXP3	action-level	paired gain	state better in
8	455.1	368.2	387.6	19.3 ± 3.4	73/100
20	618.3	493.7	547.5	53.8 ± 5.6	87/100
40	734.3	574.5	696.1	121.5 ± 9.4	99/100
80	780.4	614.8	784.0	169.3 ± 9.9	100/100

Study 1, $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, 100 paired seeds; $\pm$ is one standard error.

Observed. For the interleaved variant, sample-path pseudo-regret stayed below the bound $\sqrt{2(d+1)|\mathcal{S}|T\log K}$ in all 100 seeds at every K, that bound running from 2913 to 4228 against realized 455 to 780. The single-copy variant beat action-level weighting at every K, by a margin growing from 19 to 169 as K rose tenfold with $|\mathcal{S}|$ fixed, and beat the interleaved variant throughout. Sweeping the delay at $K = 40$ gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at $d = 10$, 50 and 200.

Interpretation. A margin that grows in K while $|\mathcal{S}|$ is held fixed is the prediction that separates the two estimators, since only the action-level cost scales with K, so the improvement is attributable to the state channel. Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap as the delay grows is what the additive delay term of Theorem 1 predicts and the multiplicative one of the interleaved bound does not.

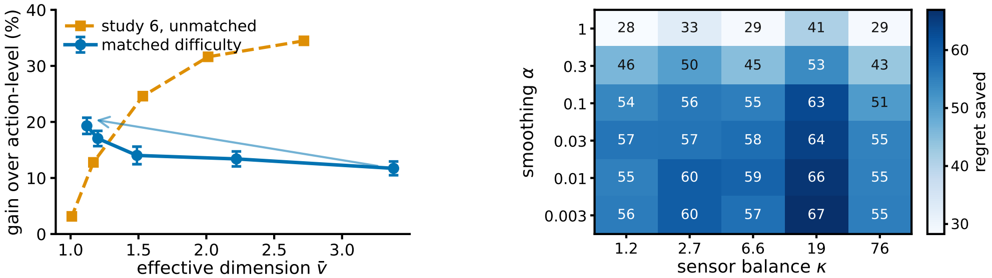

Figure 3: **Left,** study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, supporting the interpretation that the decline in the unmatched sweep is driven by task-difficulty confounding rather than by the pooling mechanism itself. Bars are one standard error of the paired difference. **Right,** regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell, from the plug-in sensitivity study whose script is released with the rest.

Limitation. This is one synthetic family with a fixed drift band, and a simulation cannot establish that a bound holds, only that sample-path pseudo-regret stayed below it in the runs performed.

Study 10. Overlap drives the gain, not easier tasks. Sweeping the overlap without further control finds the advantage falling as $\widehat{\bar{v}}$ falls, which read at face value contradicts Theorem 1, whose budget V^- falls with $\bar{v}$. That sweep is confounded, since raising the overlap also flattens $c_t = P\theta_t$ across actions, collapsing the comparator gap so that both regrets vanish together. This study holds $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$ and the difficulty fixed, and varies only $\widehat{\bar{v}}$. Difficulty is the uniform-play regret $\sum_t \mathrm{mean}_a c_t(a) - \min_a \sum_t c_t(a)$, a property of the instance with no algorithm in it, and the amplitude of θ is bisected in every cell to hit a common target. The map keeps one contrast direction alive at every overlap and θ is aligned with it, which is what lets the gap survive as the rows agree.

Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at $\widehat{\bar{v}} = 3.380$ to 19.3 per cent at $\widehat{\bar{v}} = 1.120$, with absolute gains 85.5 ± 8.9 to 106.2 ± 7.9 and a correlation of -0.929 between $\widehat{\bar{v}}$ and the gain. The unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range (Figure 3), so the apparent contradiction is an artefact of the confound rather than of the mechanism.

Interpretation. The two sweeps disagree in sign, and the matched-difficulty sweep agrees with the overlap dependence Theorem 1 predicts for the single-copy algorithm this experiment runs. Overlap is what pooling exploits, and the unmatched sweep measures overlap confounded with an instance becoming trivial.

Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller gap should if anything shrink the advantage, the measured rise is conservative in that direction. The matched gains are also smaller in magnitude than the unmatched ones, which is what fixing the difficulty costs.

Study 14/16. A recommendation funnel, and a rate-optimal baseline on it. Every other study draws P from a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an environment to the shape of a recommendation funnel instead. Actions are $K = 200$ catalogue items, states are $|\mathcal{S}| = 6$ ordered engagement depths, and the outcome is a delayed conversion. Three structural facts are imposed because funnels have them, namely that items fall into a modest number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the loss falls with engagement depth and drifts seasonally rather than adversarially. The generated catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest. The same environments, seeds and delay convention then carry a comparison against the hybrid Tsallis-plus-entropy FTRL of Zimmert and Seldin (2020), which attains the optimal rate for delayed feedback, its delay term being additive rather than multiplicative. The algorithms published since

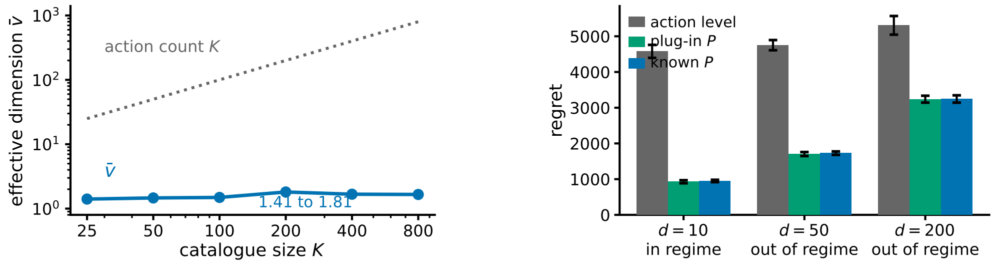

Figure 4: Study 14/16, a recommendation funnel. **Left,** the effective dimension stays near two while the catalogue grows thirty-twofold, so the gap between K and $\widehat{\bar{v}}$ comes from the catalogue having categories, not from a chosen K. **Right,** the three arms at $T = 20000$ and 8 paired seeds, labelled by whether the regime condition of Section 3 holds. Bars are one standard error. No real data is used anywhere in this study.

match that adversarial rate rather than improve on it, adding adaptivity to stochastic instances instead (Schlisselberg et al., 2025); all of them read the action alone.

Observed. The effective dimension is $\widehat{\bar{v}} = 1.97$ against $K = 200$, a hundredfold gap, and it stays between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 4). Against action-level weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at $d = 10$, 50 and 200, the largest margins in this appendix, and the plug-in matches known P at every delay, the paired difference never reaching two standard errors.

family, cell	State-EXP3	Zimmert and Seldin	same, grid-best	State-EXP3, grid-best
funnel, $d = 10$	949.0	2959.4	2413.8	27.0
funnel, $d = 50$	1728.1	3557.0	2556.4	95.9
funnel, $d = 200$	3244.8	4734.7	2807.7	199.7
overlap 0.00	661.5	731.8	610.4	202.4
overlap 0.65	584.1	673.1	544.0	51.2
overlap 0.95	461.4	540.2	419.4	34.3

Study 14/16. Regret under three tunings, on the funnel and on study 10's matched sweep; grid-best columns are chosen by reading these benchmarks.

Against the rate-optimal baseline, with State-EXP3 at the single tuning used everywhere else here and Zimmert and Seldin (2020) at the best of a six-point grid around its own value, State-EXP3 wins every cell, cutting regret by 31.5 to 67.9 per cent on the funnel and 9.6 to 14.6 per cent on the matched sweep of study 10. With both at their grid-best it wins every cell again and by more, 66.8 to 98.9 per cent. With Zimmert and Seldin (2020) alone at its grid-best it wins four of six.

Interpretation. The separation between K and $\widehat{\bar{v}}$ arrives from catalogue structure rather than from a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet families cannot make. Where the state-pooled variant is not held to the worse tuning of the two, the state channel helps against a rate-optimal method, which is expected, since that method cannot read the state at all; grid tuning helps it far more than it helps Zimmert and Seldin (2020), consistent with pooling cutting estimator variance and lower variance admitting a larger learning rate. The condition under which interleaving's bound would improve on the action-level rate fails at $d = 50$ and $d = 200$, while the single-copy variant still saves most of the regret there, which is the regime Theorem 1 was proved for.

Limitation. *No real data is used.* There is no interaction log in this environment, nothing here is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The structure is imposed by hand from qualitative properties, so it shows that the regime is describable and self-consistent, not that any deployed system occupies it. The four cells of six lost when Zimmert and Seldin (2020) alone is grid-tuned are a real loss and reported as one. No column is a symmetric protocol: the two grids are finite and unequal, and State-EXP3's grid-best sits at its top multiplier in every cell, so its own minimum may lie outside the range searched. Every grid-best multiplier is

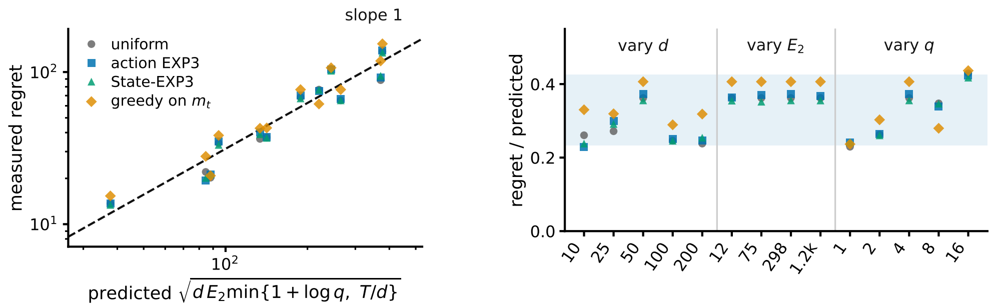

Figure 5: Study 8, the drift channel. **Left,** measured regret against the scale Theorem 2 predicts, over twelve cells in which d varies 20-fold, E_2 100-fold and J 16-fold. The dashed line has slope one. **Right,** the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

chosen by reading these benchmarks, so only the first column is a procedure a practitioner could run. Eight paired seeds on the funnel and ten on the matched sweep. The arms run $m = 1$, covered by Theorem 1, and the environment has no adversarial component, so it exercises the pooling channel and not the drift channel.

A heuristic baseline. The best-state rule identifies the state with the lowest mean loss and plays the action most likely to reach it, ignoring every other state. Across nine settings, five families at $d = 10$ and 50 plus one drift cell, it leads seven and STATE-EXP3 two. Its two losses are the spread family, built so that the best action mixes over several good states while a decoy action puts more mass than any other on the single best one: the rule takes the decoy and records 1496 ± 329 and 1495 ± 327 at $d = 10$ and 50 against our 486 ± 11 and 814 ± 20. Sweeping T from 2500 to 40000 on that family and fitting regret to a power of T gives an exponent of 1.00 for the rule, 0.58 for STATE-EXP3 and 0.67 for action-level weighting, so the rule's regret grows linearly where both learners' grow sublinearly. Its spread across seeds averages 4.6 times ours over the nine settings. The rule needs one state to be both best and reachable, and nothing in it detects when that fails, which is why the gain it shows on seven settings is not a gain a practitioner can rely on.

Study 8. The drift bound describes the right scaling. The construction of Theorem 2 at $K = 40$, $T = 8000$, $h = d$ and 12 paired seeds, sweeping d from 10 to 200, the amplitude ε from 0.02 to 0.20 so that E_2 runs from 12 to 1193, and the drifting coordinates J from 1 to 16. We run four policies: uniform, action-level EXP3, STATE-EXP3, and the greedy policy on the exact stale vector m_t, the last because Theorem 2 claims to survive it. E_2 is measured on the realized instance rather than predicted, so the normalization uses the quantity the theorem uses.

Observed. The predicted scale $\sqrt{d\,E_2 \min\{1 + \log J, T/d\}}$ ranges from 38 to 377 across the grid and the measured regret tracks it (Figure 5). Normalized, the four policies read 0.230 to 0.429, 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and 0.310 for the three that never see m_t.

Interpretation. The three arguments enter the predicted scale differently and the collapse holds in all three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data. Greedy play on the exact m_t sits at the top of the band rather than below it, which is what the bound surviving the oracle amounts to in these runs.

Limitation. Every cell has $T/d > 1 + \log J$, so only the $1 + \log J$ branch of the minimum is exercised and the T/d saturation branch, which needs d comparable to T, is not probed. A lower bound quantifies over all algorithms and four policies cannot establish that. What these runs check is the scaling of the construction.

Code and environment. The code producing every reported number and figure is at `https://github.com/melikabaghi/state-exp3`. It pins the software environment used for the reported numbers, python 3.14.3 with numpy 2.4.2 and matplotlib 3.10.8, states the seed count for each study, lists one command per numbered study, and commits the stored outputs each figure reads so that every figure rebuilds without rerunning an experiment.

Taken together these studies cover both channels. Studies 1, 10 and 14/16 probe the state channel and are consistent with the effective dimension, rather than $|\mathcal{S}|$ or K, governing what pooling buys; study 10 had to control for task difficulty before an overlap sweep pointed that way. The funnel is the only environment whose matrix is not drawn from a Dirichlet and it reports the largest margins here, on a structure imposed by hand rather than measured. It also places STATE-EXP3 against a rate-optimal delayed-bandit method: ahead in every cell when both are tuned alike, behind in four of six when that method alone is grid-tuned. Study 8 turns to the other channel and finds the drift scale of Theorem 2 tracking the data, including for a learner handed the stale losses, which is the cost no amount of pooling reaches.

F Proof of Theorem 1: additive delay without interleaving

Interleaving cannot give an additive bound. Copy i, which absorbs a share V_i of the budget $V = \sum_i V_i$, has regret $\sqrt{2V_i \log K}$, and Cauchy–Schwarz over the m copies gives $\sqrt{2mV \log K}$, tight when the V_i agree, so the $\sqrt{m}$ in Theorem 3 is not an artifact of the analysis. An additive bound has to come from a single copy.

Throughout, $L_t = \sum_{r \le t-d-1} \hat{c}_r$ and $x_t \propto e^{-\eta L_t}$, so $L_{t+1} - L_t = \hat{c}_{t-d}$, with $\hat{c}_r = 0$ for $r < 1$. The filtration $\mathcal{F}_t$ is the environment's, including outcomes not yet delivered, as in Proposition 1; under it x_t is $\mathcal{F}_{t-d-1}$-measurable, so x_{r+d} is $\mathcal{F}_{r-1}$-measurable.

The drift lemma. The natural route prices the change of measure from x_r to x_{r+d} through the normaliser Z_r, whose reciprocal has no uniformly bounded moment. That is avoidable. The pooled estimate satisfies

$$\langle x_t, \hat{c}_t \rangle = \sum_a x_t(a) \frac{P(S_t \mid a) X_t}{q_t(S_t)} = X_t \le 1$$

identically, on every path, because the denominator is the state marginal of the same distribution that supplies the numerator.

Lemma 3. *If $\eta \le 1/(e(d+1))$ then $x_{t+1}(a) \le (1+1/d)\, x_t(a)$ for every t and every a, surely, and hence $x_{t+d}(a) \le e\, x_t(a)$.*

Proof. Strong induction on t. For $t \le d$ nothing is delivered and $x_{t+1} = x_t$. For $t > d$, assume the claim for every $s < t$ and compose it over the d steps from $t-d$ to t, giving $x_t(a) \le (1+1/d)^d x_{t-d}(a) \le e\, x_{t-d}(a)$. Contracting with the fixed nonnegative vector $\hat{c}_{t-d}$ and using the identity above, $\langle x_t, \hat{c}_{t-d} \rangle \le e \langle x_{t-d}, \hat{c}_{t-d} \rangle = e X_{t-d} \le e$. With $Z_t = \sum_b x_t(b) e^{-\eta \hat{c}_{t-d}(b)}$ and $e^{-z} \ge 1 - z$, this gives $Z_t \ge 1 - \eta \langle x_t, \hat{c}_{t-d} \rangle \ge 1 - \eta e \ge d/(d+1)$, so $x_{t+1}(a) = x_t(a) e^{-\eta \hat{c}_{t-d}(a)} / Z_t \le x_t(a)/Z_t \le (1+1/d) x_t(a)$. □

What the induction bounds is $\langle x_t, \hat{c}_{t-d} \rangle = X_{t-d}\, q_t(S_{t-d}) / q_{t-d}(S_{t-d})$, the state-marginal ratio that the $1/Z_r$ route could not reach. Individual $\hat{c}_r(a)$ remain unbounded; only their x_t-weighted mass is controlled. Cesa-Bianchi et al. (2019) prove the action-level version under $\eta \le 1/(Ke(d+1))$ and Thune et al. (2019) remove the K, using that their estimate is carried by the single action played, so the weight cancels into a same-action ratio. A pooled estimate is spread over every action and does not collapse that way; the identity above replaces the cancellation.

Centring. Let $\zeta_t = \hat{c}_t - X_t \mathbf{1}$, with $\zeta_r = 0$ for $r < 1$. Exponential weights is unchanged by a shift that is constant across actions, so x_t is the same iterate and this is a change of analysis, not of algorithm (Eldowa et al., 2024). Keeping $\gamma(s) = \mathbb{E}[X_t^2 \mid S_t = s]$ inside the state sum,

$$\mathbb{E}\big[\langle x_t, \zeta_t^2 \rangle \mid \mathcal{F}_{t-1}\big] = \sum_s \gamma(s) \Big[\frac{\sum_a x_t(a) P(s \mid a)^2}{q_t(s)} - q_t(s) \Big] \le v_t - 1,$$

each bracket being nonnegative by Jensen and $\gamma(s) \le 1$, the final step being Remark 1. Bounding $\langle x_t, \hat{c}_t^2 \rangle$ by v_t first and then subtracting does not work: it leaves $v_t - \mathbb{E}[X_t^2]$.

Proof of Theorem 1. Since $\zeta_r(a) \geq -X_r \geq -1$ we have $\eta\zeta_r(a) \geq -\eta$, and Lagrange's remainder gives $e^{-z} \leq 1 - z + (e^{\eta}/2)z^2$ for $z \geq -\eta$. With $\log(1+y) \leq y$, the potential $W_t = \sum_a \exp(-\eta \sum_{r \leq t-d-1} \zeta_r(a))$ obeys $\log(W_{t+1}/W_t) \leq -\eta\langle x_t, \zeta_{t-d}\rangle + (\eta^2 e^{\eta}/2)\langle x_t, \zeta_{t-d}^2\rangle$. Telescoping from $W_1 = K$ and using $\log W_{T+1} \geq -\eta \sum_{r \leq T-d} \zeta_r(a^*)$ for the comparator a^*,

$$\sum_{r=1}^{T-d} \langle x_{r+d}, \zeta_r\rangle \leq \sum_{r=1}^{T-d} \zeta_r(a^*) + \frac{\log K}{\eta} + \frac{\eta e^{\eta}}{2} \sum_{r=1}^{T-d} \langle x_{r+d}, \zeta_r^2\rangle.$$

The shift cancels between the two sides. As x_r is $\mathcal{F}_{r-d-1}$-measurable, $\mathbb{E}[\langle x_r, \hat{c}_r\rangle] = \mathbb{E}[\langle x_r, c_r\rangle]$ and $\mathbb{E}[\hat{c}_r(a^*)] = c_r(a^*)$, and the d rounds undelivered by $T+1$ contribute at most d, so

$$\mathbb{E}[R_T] \leq \frac{\log K}{\eta} + \frac{\eta e^{\eta}}{2} \sum_r \mathbb{E}\big[\langle x_{r+d}, \zeta_r^2\rangle\big] + \sum_r \mathbb{E}\big[\langle x_r - x_{r+d}, \hat{c}_r\rangle\big] + d.$$

Lemma 3 gives $x_{r+d} \leq e\, x_r$ surely, and x_{r+d} is $\mathcal{F}_{r-1}$-measurable, so the second sum is at most $e \sum_r \mathbb{E}[v_r - 1] \leq eV^-$.

For the third, write $x_{r+d}(a) = x_r(a) e^{-\eta G_r(a)}/Z'$ with $G_r = \sum_{u=r-d}^{r-1} \hat{c}_u \geq 0$ and normaliser $Z' \leq 1$, distinct from the one-step Z_t of the Lemma. Then $x_r(a) - x_{r+d}(a) \leq x_r(a)(1 - e^{-\eta G_r(a)}) \leq \eta x_r(a) G_r(a)$ for every a, trivially where the left side is negative. Since $\hat{c}_r \geq 0$ and x_r, G_r are $\mathcal{F}_{r-1}$-measurable, the conditional expectation is at most $\eta\langle x_r, G_r\rangle$, and $\mathbb{E}[\langle x_r, G_r\rangle] = \sum_{u=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_u\rangle] \leq d$: for each such u, x_r is $\mathcal{F}_{r-d-1}$-measurable and $r-d-1 \leq u-1$, so it leaves the conditional expectation and $\mathbb{E}[\langle x_r, \hat{c}_u\rangle] = \langle x_r, c_u\rangle \leq 1$. This is measurability, not independence, and at $u = r-d$ the two σ-fields coincide, so the step rests on the timing convention of Section 2. The sum is at most ηdT.

Finally $\eta \leq 1/(2e)$ gives $e^{1+\eta} < 3.3$, so $\mathbb{E}[R_T] \leq \log K/\eta + \frac{\eta}{2}(3.3V^- + 2dT) + d$. For $f(\eta) = A/\eta + B\eta$ restricted to $\eta \leq \eta_{\max}$ one has $\min f \leq 2\sqrt{AB} + A/\eta_{\max}$, in the interior and in the binding case alike; with $A = \log K$, $B = (3.3V^- + 2dT)/2$ and $\eta_{\max} = 1/(e(d+1))$ this is the stated bound. $\square$

The cap on η binds only when $d \gtrsim T/(e^2 \log K)$, and not at the settings of Appendix E: the funnel study's own rates are $0.0058, 0.0031$ and 0.0016 at $d = 10, 50, 200$ against caps $0.0334, 0.0072$ and 0.0018, so Theorem 1 covers those runs as they were executed. Over three seeds and 20000 rounds the induction has room to spare, $\max_a x_{t+1}(a)/x_t(a)$ reaching 1.006, 1.003 and 1.002 against the permitted 1.100, 1.020 and 1.005.

The two bounds are numerically close on the funnel, where $\bar{v} \approx 2$: the additive form carries $3.3(\bar{v} - 1) + 2d$ and the interleaved one $(d+1)\bar{v}$, and these cross near $\bar{v} = 2$. The additive form gains as $\bar{v}$ rises toward $|\mathcal{S}|$, and it describes the algorithm that is actually run, which is the reason to prefer it.